Supporting Information for "Hiding the peak, not missing the days: support-based screening of incomplete river-temperature archives"

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Introduction

This supporting information records what belongs to the record rather than to the argument: data provenance, protocol history, the predictions that were written down and not met, the runs that were discarded and why, and the reproduction path. Nothing here is required to follow the article.

S1 Data sources

Daily water-temperature series were assembled from nine public providers (Table S1); the external panel adds France's Hub'Eau and the UK Environment Agency. Each provider's product, documented statistic, quality codes, licence and acquisition route were re-verified against the provider's own documentation on 2026-10-12 (`results/wrr_submission_ready/measurement_provenance_dictionary.csv`). Eight of the eleven publish under a named open licence and none required a data-use agreement, but LUBW reserves copyright on its raw values: those seven networks are not redistributed here and are read locally through LUBW's own service.

Station coordinates were assembled for all 457 stations: from the providers' own catalogues already held in the repository for FOEN, CHMI, SYKE, LUBW, GKD Bayern, NVE and Rijkswaterstaat and for nine USGS gauges; from the USGS

Table S1. Providers and their role in the study.

provider	country	role
ARSO	Slovenia	development
CHMI	Czechia	development
FOEN	Switzerland	development
GKD Bayern	Germany (Bavaria)	development
LUBW	Germany (Baden-Württemberg)	development
NVE	Norway	development
Rijkswaterstaat	Netherlands	development
SYKE	Finland	development
USGS	United States	development and external
Hub'Eau	France	external
UK Environment Agency	United Kingdom	external

NWIS site service for the remaining United States gauges; and from ARSO's public station feed. Every row of the coordinate table in the repository records its source.

Daily 2 m air temperature is NASA POWER (MERRA-2), 1 January 1981 to 31 August 2026, one point per station at the product's native resolution. Basin characteristics for the United States subset are GAGES-II, joined on the eight-digit gauge identifier.

S2 Quality control and the definition of a network

A *network* is a provider's group of stations on a common river system, which is the unit at which donor stations are available to one another. Networks are the clustering unit for every interval in the article, because station-years within a network share weather, regulation and instrumentation. Two labels that share a station are merged into one *independence group* before any fold is drawn, which is why 49 external labels form 41 validation groups.

Waterbody class was assigned per network from the provider's own naming and an explicit map; seas, lakes, canals, estuaries and deltas are listed and excluded rather than averaged into a river result. Reference series are never imputed: a calen-

dar year with any missing day contributes no truth. Stations with fewer than four fully observed calendar years are excluded, because the leave-one-year-out estimator protocol requires them. The same stations contribute 2,422 station-years with at least 360 daily slots and at least one missing day; those form the real-calendar population of Section 4.4 of the article.

S3 Protocol history

The study was pre-registered before the reserved networks were opened on 2026-09-23, and several rounds of repair followed. Three early repairs are recorded here rather than silently applied.

The incumbent's imputed limit was mis-stated. An earlier version placed the comparison at a thirteen-week (91-day) limit attributed to Johnson et al. (2021). The source says twenty weeks. The analysis code now uses 140 days.

The weighting between outage arms was not declared. The case set mixes an empirical arm, drawn from the outage geometry the same networks experience, with a stratified arm built to span the decision space. Their base rates differ by an order of magnitude. The pre-registered test was run on the pooled set, as declared, and the decomposition is reported in the earlier development round.

The natural scale of the peak has two values. Computed over the full eligible roster it is 1.06 °C; recomputed inside the development case table it is 1.08 °C. The primary tolerance is 0.50 °C throughout; the tolerance sweep shows the axis ordering is unchanged between 0.25 and 1.0 standard deviations.

The 2026-10-19 round repaired two label-isolation defects: outer folds now keep their own calibration scores, and networks that share a station are kept in one folding unit. The 2026-10-20 round repaired the object mismatch and the tie rule that this supporting information documents below. The round's analyses are registered under `major_revision_20261020/`: the aligned admission (`p0_01_aligned_policy_admission`), the selector calibration (`p0_02_selection_calibration`), the real-calendar adoption comparison (`p0_03_real_calendar_adoption`), the hot-year closure (`p0_04_hot_year_closure`), the repair attribution (`p0_05_repair_attribution`) and the figure sources (`p0_06_figures`). Each holds a `SUMMARY.md`, a `protocol_f`

`ingerprint.json` with script and input hashes and the git head, and the machine-readable tables the article cites.

S4 The object mismatch, and what it changed

The 2026-10-19 run declared the stored policy reconstruction as the admitted object, but built the failure event from the certificate's absolute error. The two are different reconstructions of the same station-year. On the external panel the policy error exceeds the tolerance in 1,996 of 13,017 cases and the certificate error in 2,078; 408 cases fail only under the policy and 490 only under the certificate, so the two events disagree on 898 cases, or 6.90% of the panel. On the development panel they disagree on 830 of 40,704 cases (2.04%; 478 policy-only, 352 certificate-only).

The aligned protocol recomputes everything—threshold selection, evaluation, intervals and contrasts—on `abserr_policy`, after checking in-script that `abserr_policy` = $|\texttt{hottest30}_{\text{policy choice}} - \texttt{truth_hottest30}|$ on every finite row. The certificate bounds survive only as ranking scores.

Table S2 reports the 2×2 consequence of the two repairs. On the external panel the fitted-score advantage over the peak-window rule is 2.7 points when the event is the certificate and the tie rule admits whole blocks (2.8 with jitter), and 7.2 points when the event is the policy reconstruction and ties are admitted whole. Most of the change (about 4.5 points) comes from measuring the right object; the tie rule contributes about 1.3 points. On the development panel the two repairs move the same contrast by less than one point, which is why the defect was easy to miss without an external comparison.

S5 Measurement contract and coverage

Every result is reported against the daily-value statistic its providers actually publish (Table S4). Of the 40,704 development cases, 74.4 % come from the five providers that document a daily mean (84 networks), 15.5 % are project aggregates of sub-daily or instantaneous values (14 networks), 8.3 % come from ten networks whose provider never states whether the published daily value is a mean, and 1.7 % are Finnish morning reference readings. The external wave-2 panel is 94.4 % documented daily means. A station-level coverage table with first and last year, complete

Table S2. Retention at the 5% cap under the two failure events and the two tie rules. Same scores, same folds, same selection rule; entries are the mean of three fold seeds. "Jitter" reproduces the 2026-10-19 construction and applies it to calibration and evaluation alike.

panel	event / tie	fitted	peak-window	length	fitted − peak (pp)
external	policy / whole block	0.739	0.667	0.396	+7.2
external	policy / jitter	0.741	0.656	0.445	+8.5
external	certificate / whole block	0.728	0.701	0.406	+2.7
external	certificate / jitter	0.728	0.699	0.445	+2.8
development	policy / whole block	0.704	0.607	0.305	+9.7
development	policy / jitter	0.704	0.602	0.324	+10.2
development	certificate / whole block	0.706	0.616	0.305	+9.0
development	certificate / jitter	0.708	0.622	0.342	+8.6

years, statistic class, quality class and licence route for every station is in `major_revision_20261014/p0_06_coverage_recompute/final_coverage_table.csv`.

The aligned rules were re-selected and re-evaluated inside the documented-daily-mean subset (Table S3). Thresholds were chosen on the subset's calibration rows only and the stored per-fold calibration arrays were restricted the same way; nothing was refitted. The fitted score's advantage over the peak-window rule is +8.0, +7.3 and +7.3 points on the external subset and +8.8, +8.7 and +10.4 on the development subset, with every interval excluding zero, so the ranking reported in the article does not depend on the project-aggregate or undocumented providers. Its comparison with the certificate bound stays interval-crossing on development (−2.3 to −4.2 points) and is positive but mostly small on the external subset (+0.1 to +1.9).

S6 Parameters and their roles

Every quantity that enters a result has one of four roles, and mixing them is a common source of confusion (Table S5).

Software and data are archived as follows. The analysis code, the frozen configuration, the derived tables and the figure sources are in the accompanying repository; the study inputs are packaged for confidential review (`dist/wrr_study_inp`

Table S3. The aligned rules re-selected inside the documented-daily-mean subset (30,289 development cases, 12,282 external cases; shares of the full panel 0.744 and 0.944). Retention is relative to the subset. Contrasts are paired, group-clustered and reported as fitted minus the named rule in points.

panel	rule	retention	failure rate	vs peak (pp)	vs bound (pp)
external	fitted risk	0.749	0.031	+7.5	+1.0
external	peak-window	0.674	0.032	—	—
external	certificate bound	0.739	0.032	—	—
external	gap-length	0.411	0.018	—	—
development	fitted risk	0.733	0.030	+9.3	−3.0
development	peak-window	0.640	0.037	—	—
development	certificate bound	0.764	0.036	—	—
development	gap-length	0.347	0.019	—	—

`uts_v1.tar.gz`) and a permanent archive with a DOI is deposited on acceptance. Restricted material is listed rather than shipped: LUBW raw values are read locally and not redistributed, and providers whose terms name no licence are used under attribution only (`metadata/data_rights.csv`). Input-file hashes for every network are in `current_data_manifest.csv`, and each analysis writes a protocol fingerprint holding the script hash, the input hashes and the git head.

S7 Retention and hot-year composition in one place

Table S23 collects the retention and realised risk of every rule on both designed panels, at the same 5% cap, for readers who want the decision table without the figure. The reference arm is included for scale: on the designed panels it admits everything and therefore fails the cap by a wide margin, which is why it does not appear in the article's panel (a).

Table S24 gives the hot-year composition behind Figure 5c. Under the pre-declared unique-station-year weighting the fitted score's total deviation is +0.9 points ($[0.0, 1.8]$) rather than +0.7, the ordering of the rules is unchanged, and the gap-length rule's refusal of true hot years remains the largest single effect in the table,

Table S4. Daily-value statistic by provider, as documented by the provider, with the class assigned to each panel case.

provider	statistic documented	stations	class used in the analysis	evidence
USGS	daily mean	427	documented daily mean	NWIS daily-values service; parameter 00010
GKD Bayern	daily mean	55	documented daily mean	provider daily series documentation
LUBW	daily mean	22	documented daily mean	provider daily series documentation
NVE	daily mean	57	documented daily mean	HydAPI daily values
FOEN	daily mean	11	documented daily mean	opendata.swiss daily series
CHMI	sub-daily (10-minute)	73	project aggregate	provider sub-daily series
Rijkswaterstaat	raw measurements	5	project aggregate	WaterWebServices raw readings
Hub'Eau	continuous	9	project aggregate	Hub'Eau chronique
ARSO	daily values, statistic unstated	47	undocumented	provider page only
SYKE	08:00 near-shore reading	3	non-daily reference reading	provider documentation

aCounts are station-network rows from the coverage table; a station appearing in two networks is counted once per network. The class assignment is a deterministic parse of `measurement_provenance_dictionary.csv`.

Table S5. Model and decision parameters by role. "Fitted" means estimated from data at run time; "fixed" means a constant chosen before any result was seen; "calibrated" means chosen on groups disjoint from the cases it is applied to; "declared" means a study tolerance rather than a model quantity.

quantity	role	value or source
ridge penalty of the donor and air estimators	fixed	$\alpha = 1.0$, seven air lags, whole-year fit
gradient-boosted risk score	fitted	quantile 0.9, depth 4, 350 iterations, learning rate 0.06, leaf 40; refit per outer fold
risk-score target	declared	absolute error of the admitted policy reconstruction
admission threshold	calibrated	largest candidate whose independence-group clustered bootstrap percentile at $1 - 0.05/n_{\text{cand}}$ stays under 5 %
tie rule	declared	no jitter; a threshold is an observed score and $s \leq \tau$ admits whole tie blocks
consequential-error tolerance τ	declared	0.5 °C
fold unit	fixed	connected component of networks that share a station id
folds	fixed	5 outer folds, 3 fold seeds (101/202/303), 4 inner folds for the risk score
bootstrap draws	fixed	1,000 for threshold selection, 2,000 for evaluation intervals, 10,000 for the stability check

but that is its retention and not a selection against hot years: against the retention of all records the excess refusal of true hot years is negative for every rule on the external panel (Table S21). The case-weighted estimate is primary because it is the archive an agency holds under this monitoring design.

S8 The selector's risk control, read correctly

The deployed threshold reads the percentile $1 - 0.05/n_{\text{cand}}$ of an independence-group clustered bootstrap of the calibration false-admit rate, with n_{cand} the number of distinct snapped thresholds. Two properties were separated in simulation on 2,000 synthetic clustered archives per scenario, with the deployed rule imported from the analysis code so the procedure cannot drift:

- the *population* risk of the selected threshold, measured on an independent two-million-case reference draw from the same generator; and
- the error rate observed in a finite fresh test batch of 16 groups.

Table S6 reports both. With the deployed simultaneous bound, the selected threshold's population risk exceeds the 5% cap in 0.4% of repetitions in the baseline scenario, 0.6% near the cap, 0.4% with unequal group sizes and 2.4% with twelve calibration groups. The finite test batch exceeds the cap far more often (7.8%, 7.5%, 4.6% and 12.2%) because the batch adds noise the rule cannot control. The 8.5% quoted in the previous version of this work was the second quantity; reading it as the selector's error rate was a misinterpretation. A pointwise bound keeps about twice as many records (35.9% against 18.5% in the baseline scenario) and violates the cap about six times as often (7.4% against 0.4% on the population measure).

The extreme percentile is numerically delicate at 1,000 draws. Re-running the selection with 10,000 draws changes the chosen threshold in 42 of 150 fold-by-score cells. The fitted score's retention moves by at most 1.0 point and the discrete rules by at most 3.9 points. The article therefore describes the selected threshold as a calibration decision at this granularity, not as a stable optimum.

Table S6. Selector calibration per scenario, 2,000 repetitions each. "True risk" is the population failure rate of the selected threshold on an independent two-million-case reference draw; "batch" is the rate observed in a finite fresh test sample.

scenario	procedure	retention	true risk	P(true > 5%)	P(batch > 5%)	batch rate
baseline	deployed	0.185	0.0199	0.004	0.078	0.0204
baseline	pointwise	0.359	0.0352	0.074	0.191	0.0356
low error	deployed	0.761	0.0313	0.000	0.051	0.0320
low error	pointwise	0.881	0.0386	0.000	0.153	0.0394
near cap	deployed	0.288	0.0222	0.006	0.075	0.0224
near cap	pointwise	0.543	0.0365	0.066	0.200	0.0369
unequal groups	deployed	0.296	0.0306	0.004	0.046	0.0308
unequal groups	pointwise	0.425	0.0406	0.064	0.181	0.0408
few groups	deployed	0.153	0.0171	0.024	0.122	0.0173
few groups	pointwise	0.309	0.0311	0.119	0.254	0.0322

S9 Predictions that failed, and one that was withdrawn

The prediction that failed this round. Before the aligned rerun we predicted, on the basis of the 2026-10-19 numbers, that the fitted score would not clear the pre-specified five-point margin against the peak-window rule even after the object was repaired. It clears it on the external panel at all three fold seeds (point estimates 8.0, 7.3 and 7.0 points, lower interval limits 5.0, 4.6 and 4.3) and on the development panel (9.7 to 10.3 points). The earlier reading was an artefact of the object mismatch, and we report the corrected result rather than the prediction.

A justification that did not survive the ladder. The article's earlier account of the fitted score's gain was that it reads something the support share does not contain. That is true, but the 2026-10-14 same-information ablation had already found that adding the support share to a generic model changes retention by 0.00 points ($[-1.00, +1.00]$), and the manuscript did not carry it. The aligned protocol reproduces it (Table 3 of the article and Tables S11 and S12 here): the support share adds $+0.3$ to $+0.8$ points on the external panel and -0.5 to $+0.5$ on the develop-

ment panel, and the gain over the peak-window rule belongs to donor information and reconstruction diagnostics.

A claim withdrawn rather than falsified. An earlier version framed the result as "position, not length, governs". The crossed design does not support a law: the support effect is universal in sign but its size ranges from 0.24 to 0.64 °C at 30 days across reconstructors, and linear interpolation retains a large length effect at a high hidden share (0.82 °C [0.64, 1.00] between 90 and 30 days at a true share of at least 0.9). The other four reconstructors keep a small length effect, 0.05 to 0.14 °C, that is distinguishable from zero (Table S17). The article now states the conditional version.

Three earlier predictions that were not met remain in the record from the development round: the support-restricted advantage does not decay monotonically in support width; the estimand's own support is not the best conditioning set at every width; and two hand-chosen phase windows do not beat both peak overlap and outage length.

S10 Runs that were discarded

Five defects made specific artifacts unusable, and each was repaired before the numbers reported here:

1. outage realisations were once seeded from a process-salted hash, so the case table could not be regenerated;
2. worker results were once concatenated in completion order, so row order was not reproducible;
3. an additive per-estimator conformal scheme under-covered, because selecting the smallest of several bounds is a winner's curse that per-estimator calibration cannot see;
4. a shared calibration-score array let a later outer fold overwrite earlier folds' scores with models that had seen those folds' labels (repaired 2026-10-19);
5. the failure event was read from the certificate reconstruction while the admitted object was the policy reconstruction, and the two tie rules differed between calibration and deployment (repaired 2026-10-20).

In each case the affected numbers were discarded rather than reported alongside the repaired ones.

S11 The attribution round: design audit, ladder, channels and rule

This section holds the tables behind the article's attribution claims. They are written by `scripts/346_wrr_si_tables_r2.py` from the artifacts of `scripts/340` to `scripts/344` under `results/wrr_submission_ready/major_revision_20261020b/`, and nothing in them is typed by hand.

What the designs contain. The earlier article described the 53,721 designed outages as a factorial design with scattered extra missing days. The case tables say otherwise, and Tables S7 to S9 give the corrected description: single contiguous runs of 14, 30, 60, 90, 120, 180 or 270 days (stratified) and transplanted real geometries (empirical); no scattered days; the overlap class defined on the climatological window; and a correlation of gap length with the true hidden share of +0.52 that the design does not remove. The crossed panel (1,899 outages) was generated separately on 24 development stations; it also holds 19 transplanted outages, which no analysis here uses.

The ladder. Tables S10, S11 and S12 give the retention and the paired contrasts of every specification of the information ladder, including the two non-deployable specifications that read the true hidden share.

Channels, interventions and the misalignment contrast. Tables S13 to S16.

Sensitivity of the length effect. Table S17 compares the within-station-year estimator of the article with the network-subset estimator of the earlier mechanism run; Table S18 gives the competing-peak relation.

The screening-need rule and the hot-year null. Tables S19 to S22. The real replay's evidence is two to four networks that need a screen on 3 to 6 target station-years each, and the tables are the complete record of it.

S12 Verification and reproduction

- The case tables are byte-reproducible including row order, pinned by a test.

Table S7. What the designed panels contain. *Stratified* outages are one contiguous run placed at a requested overlap with the climatological peak window; *empirical* outages transplant real gap lengths and months (one to four runs). Station-years are physical station-years (the external panel's 1,267 label-station-years are 1,103 station-years). Source: `r2_01_design_audit/arm_inventory.csv`.

panel	arm	outages	station-years	networks	median gap (d)	runs per outage	single-run (%)
development	all	40,704	4,045	110	60	1.40	76
development	empirical	14,995	3,823	102	22	2.09	36
development	stratified	25,709	4,045	110	90	1.00	100
external	all	13,017	1,103	49	30	1.49	73
external	empirical	4,977	1,103	49	11	2.28	30
external	stratified	8,040	1,103	49	90	1.00	100
crossed (mechanism)	all	1,918	211	12	60	1.01	99
crossed (mechanism)	empirical	19	13	1	30	2.11	21
crossed (mechanism)	stratified	1,899	211	12	60	1.00	100

Table S8. Correlation of gap length with hidden support in each design, with network-clustered 95% intervals (2,000 draws). Only the crossed panel breaks the correlation with the climatological overlap; the real replay has the strongest. Source: `length_support_correlation.csv`.

panel	arm	hidden support	Pearson r [95% interval]	networks
development	all	true hidden share	+0.52 [+0.47, +0.55]	110
development	all	climatological overlap	+0.45 [+0.41, +0.49]	110
development	empirical	true hidden share	+0.64 [+0.47, +0.75]	102
development	empirical	climatological overlap	+0.64 [+0.45, +0.75]	102
development	stratified	true hidden share	+0.39 [+0.37, +0.41]	110
development	stratified	climatological overlap	+0.28 [+0.26, +0.29]	110
external	all	true hidden share	+0.52 [+0.50, +0.53]	49
external	all	climatological overlap	+0.46 [+0.44, +0.49]	49
external	empirical	true hidden share	+0.57 [+0.43, +0.67]	49
external	empirical	climatological overlap	+0.56 [+0.42, +0.67]	49
external	stratified	true hidden share	+0.37 [+0.35, +0.39]	49
external	stratified	climatological overlap	+0.27 [+0.25, +0.30]	49
crossed (mechanism)	all	true hidden share	+0.14 [+0.12, +0.16]	12
crossed (mechanism)	all	climatological overlap	+0.00 [−0.02, +0.03]	12
crossed (mechanism)	stratified	true hidden share	+0.14 [+0.12, +0.16]	12
crossed (mechanism)	stratified	climatological overlap	−0.01 [−0.02, +0.00]	12
real calendar replay	all	climatological overlap	+0.71 [+0.44, +0.89]	14

Table S9. Length-by-class cells of the stratified arm that are empty or hold fewer than 100 outages. A 14-day run cannot cover a 30-day window, and a 270-day run can hardly avoid it. In the crossed panel every one of the nine cells holds 211 outages, one per station-year.

panel	target length (d)	overlap class	outages	status
development	14	full	0	empty
development	270	none	0	empty
development	270	partial	58	sparse (¡100)
external	14	full	0	empty
external	270	none	10	sparse (¡100)
external	270	partial	13	sparse (¡100)

- No feature may read a day the outage hides; enforced in the feature constructor and pinned by a test.
- The aligned admission script asserts that `abserr_policy` equals the absolute difference between the stored policy prediction and the reference value before it uses it, so an alignment error aborts the run.
- The failure event, the threshold and the evaluated error are the same column in every stage of the aligned protocol; a unit test checks that the case-level admissions reproduce the thresholds.
- Every table cell and audited inline number in the article is re-derived from the artifacts by the build script, which fails when a listed number is missing from the text.
- Every input the current study reads is listed with a byte count and a SHA-256 in `current_data_manifest.csv`, and each analysis records its own protocol fingerprint.

Table S10. Retention at the 5% cap and the failure rate among admitted records for every specification of the ladder, mean of three fold seeds (%). One learner, one training set, one nested cross-fit and one calibration procedure throughout; the full specification reproduces the fitted score of the aligned round bit for bit. Source: `ablation_operating_points.csv`.

specification	external retention	external failure	develop. retention	develop. failure
gap-length rule	39.6	1.8	30.5	2.4
peak-window rule	66.6	3.3	60.4	3.8
warm-season coverage rule	50.7	2.0	44.3	2.0
certificate bound	72.9	3.2	72.7	3.6
gap geometry and season coverage	63.2	2.9	58.7	3.4
gap and season + peak-window share	66.7	2.9	60.4	3.3
+ donor information	68.7	3.1	62.4	3.1
gap, season, donors + peak-window share	71.5	3.0	63.3	2.9
+ reconstruction diagnostics	74.0	3.1	70.2	3.0
gap, season, donors, diagnostics + peak-window share	74.8	3.2	70.5	3.0
all generic features (no support)	73.4	3.1	70.4	3.0
+ peak-window share (fitted score)	73.9	3.1	70.5	3.0
small logistic, six features	62.4	3.2	46.3	2.8
small logistic + peak-window share	68.9	3.2	59.8	3.1
TRUE-hidden-share rule (not deployable)	72.0	3.2	69.1	3.5
generic + TRUE hidden share (not deployable)	75.0	2.9	73.9	3.1

Table S11. External panel: paired retention differences at the 5% cap, in points, with group-clustered 95% intervals (2,000 draws) at each fold seed. The three seeds re-split one panel. "Fitted score" is the generic model plus $\hat{h}$; "generic" is gap, season, donors, diagnostics and history; "diag." is the reconstruction diagnostics. Source: `ablation_contrasts.csv`.

contrast	seed 101	seed 202	seed 303
fitted score − generic, no $\hat{h}$	+0.4 [−0.6, +1.3]	+0.8 [−0.1, +1.8]	+0.3 [−0.4, +0.9]
gap+season+donors+diag.+$\hat{h}$ − gap+season+donors+diag.	+1.2 [+0.5, +1.9]	+0.8 [+0.2, +1.6]	+0.3 [−0.5, +1.2]
gap+season+donors+$\hat{h}$ − gap+season+donors	+3.0 [+1.6, +4.5]	+2.2 [+1.0, +3.4]	+3.3 [+1.8, +4.8]
gap+season+$\hat{h}$ − gap+season	+2.4 [+1.5, +3.4]	+5.0 [+3.9, +6.2]	+3.0 [+2.2, +4.0]
logistic+$\hat{h}$ − logistic	+6.2 [+5.1, +7.3]	+6.8 [+5.3, +8.4]	+6.6 [+5.3, +7.9]
generic, no $\hat{h}$ − peak-window rule	+7.0 [+3.7, +10.0]	+6.7 [+3.2, +9.7]	+6.9 [+3.8, +9.7]
fitted score − peak-window rule	+7.4 [+4.3, +10.4]	+7.5 [+4.9, +9.8]	+7.2 [+4.4, +9.8]
gap+season − peak-window rule	−2.0 [−3.3, −0.8]	−5.3 [−6.8, −4.1]	−2.7 [−4.0, −1.6]
logistic − peak-window rule	−3.7 [−6.4, −1.2]	−4.4 [−7.2, −1.8]	−4.5 [−7.4, −1.8]
gap+season+donors − gap+season	+4.3 [+0.0, +8.8]	+6.9 [+2.7, +11.0]	+5.1 [+0.8, +9.2]
gap+season+donors+diag. − gap+season+donors	+5.4 [+3.8, +7.1]	+5.7 [+3.7, +7.8]	+4.8 [+3.3, +6.4]
generic, no $\hat{h}$ − gap+season+donors+diag.	−0.7 [−1.9, +0.3]	−0.7 [−1.9, +0.6]	−0.2 [−1.5, +0.7]
generic, no $\hat{h}$ − certificate bound	+0.6 [−1.2, +2.5]	+0.4 [−1.5, +2.3]	+0.6 [−1.3, +2.2]
fitted score − certificate bound	+1.0 [−0.3, +2.3]	+1.2 [−0.5, +2.7]	+0.8 [−1.0, +2.4]
true-share rule − peak-window rule	+5.9 [+5.1, +6.8]	+5.0 [+4.0, +6.1]	+5.5 [+4.6, +6.4]
generic+true share − fitted score	+1.8 [+0.8, +2.8]	+0.8 [−0.1, +1.7]	+0.6 [−0.2, +1.4]

Table S12. Development panel: paired retention differences at the 5% cap, in points, with group-clustered 95% intervals (2,000 draws) at each fold seed. The three seeds re-split one panel. "Fitted score" is the generic model plus $\hat{h}$; "generic" is gap, season, donors, diagnostics and history; "diag." is the reconstruction diagnostics. Source: `ablation_contrasts.csv`.

contrast	seed 101	seed 202	seed 303
fitted score − generic, no $\hat{h}$	−0.5 [−1.1, −0.0]	+0.4 [−0.1, +0.9]	+0.5 [+0.1, +0.9]
gap+season+donors+diag.+$\hat{h}$ − gap+season+donors+diag.	+0.6 [+0.1, +1.1]	−0.6 [−1.1, −0.2]	+1.1 [+0.7, +1.5]
gap+season+donors+$\hat{h}$ − gap+season+donors	+1.3 [+0.4, +2.2]	+0.7 [−0.5, +2.0]	+0.6 [−0.2, +1.5]
gap+season+$\hat{h}$ − gap+season	+2.7 [+1.4, +4.2]	+1.3 [+0.1, +2.7]	+1.3 [+0.1, +2.4]
logistic+$\hat{h}$ − logistic	+11.4 [+8.7, +14.5]	+16.3 [+11.6, +22.1]	+12.8 [+7.8, +18.8]
generic, no $\hat{h}$ − peak-window rule	+10.7 [+8.2, +13.7]	+9.8 [+6.9, +13.1]	+9.3 [+6.8, +12.4]
fitted score − peak-window rule	+10.2 [+7.4, +13.4]	+10.3 [+7.2, +13.6]	+9.8 [+7.1, +13.0]
gap+season − peak-window rule	−2.1 [−3.7, −0.6]	−1.2 [−3.0, +0.3]	−2.0 [−3.5, −0.6]
logistic − peak-window rule	−10.8 [−15.3, −6.5]	−18.3 [−26.3, −11.1]	−13.1 [−21.2, −5.7]
gap+season+donors − gap+season	+3.2 [+0.4, +6.5]	+3.2 [+0.2, +6.7]	+4.9 [+1.7, +8.3]
gap+season+donors+diag. − gap+season+donors	+9.0 [+7.1, +11.0]	+8.2 [+6.2, +10.4]	+6.1 [+4.1, +8.4]
generic, no $\hat{h}$ − gap+season+donors+diag.	+0.6 [−0.2, +1.2]	−0.4 [−1.0, +0.4]	+0.4 [−0.4, +1.1]
generic, no $\hat{h}$ − certificate bound	−2.1 [−5.3, +0.9]	−1.9 [−5.5, +1.6]	−2.9 [−6.3, +0.5]
fitted score − certificate bound	−2.7 [−6.0, +0.5]	−1.5 [−5.0, +2.0]	−2.4 [−5.8, +0.9]
true-share rule − peak-window rule	+9.3 [+8.5, +10.1]	+8.2 [+7.5, +9.0]	+8.7 [+8.0, +9.5]
generic+true share − fitted score	+4.1 [+2.9, +5.3]	+3.4 [+2.4, +4.3]	+2.8 [+1.9, +3.6]

Table S13. The hiding channel $\bar{e}_S$ and the false-peak channel F of the identity in Section 3.5 of the article, by reconstructor and true hidden share of the peak window, crossed panel (°C; means over outages). The identity holds to 2×10^{-13} °C on all 9,449 finite cases. The last column is the share of outages whose false peak exceeds 0.1 °C. Source: `r2_03_three_windows/channels_by_hidden_class.csv`.

reconstructor	hidden share	outages	mean |error|	hiding	false peak	false peak > 0.1 (%)
climatology	0	772	0.060	+0.000	+0.060	9
climatology	(0, 1)	749	0.589	−0.362	+0.316	57
climatology	1	378	0.918	−0.751	+0.433	72
smoothed	0	772	0.056	+0.000	+0.056	8
smoothed	(0, 1)	749	0.561	−0.410	+0.291	54
smoothed	1	378	0.923	−0.811	+0.376	59
interpolation	0	772	0.012	+0.000	+0.012	3
interpolation	(0, 1)	749	0.374	−0.316	+0.189	43
interpolation	1	378	1.256	−2.232	+1.023	93
donor	0	750	0.017	+0.000	+0.017	4
donor	(0, 1)	736	0.252	−0.085	+0.060	16
donor	1	370	0.437	−0.172	+0.063	15
air	0	769	0.025	+0.000	+0.025	5
air	(0, 1)	749	0.469	−0.395	+0.169	35
air	1	378	0.848	−0.712	+0.176	37

Table S14. Interventions on the gap residuals, crossed panel, change in mean absolute error (°C) with network-clustered 95% intervals. Numerical experiments on recorded residuals; truth, gap and reconstructor are held fixed. Source: `residual_interventions.csv`.

reconstructor	recorded	residuals permuted	gap mean removed	gap mean only
climatology	0.440	−0.085 [−0.117, −0.055]	−0.241 [−0.299, −0.190]	−0.099 [−0.133, −0.067]
smoothed	0.428	−0.083 [−0.114, −0.055]	−0.222 [−0.272, −0.174]	−0.095 [−0.128, −0.066]
interpolation	0.402	−0.050 [−0.072, −0.029]	−0.198 [−0.242, −0.158]	−0.044 [−0.066, −0.023]
donor	0.194	−0.040 [−0.054, −0.027]	−0.114 [−0.148, −0.076]	−0.043 [−0.056, −0.030]
air	0.365	−0.097 [−0.156, −0.060]	−0.192 [−0.231, −0.152]	−0.099 [−0.153, −0.060]

Table S15. Misaligned minus aligned mean absolute error (°C), pooled and within strata of the climatological class and the true class (weights are the strata's case counts), with network-clustered 95% intervals. Source: `misalignment_within_strata.csv`.

reconstructor	pooled	within climatological class × true class
climatology	+0.66 [+0.52, +0.79]	+0.19 [+0.07, +0.26]
smoothed	+0.63 [+0.52, +0.74]	+0.23 [+0.11, +0.30]
interpolation	+0.66 [+0.54, +0.79]	+0.13 [+0.07, +0.17]
donor	+0.31 [+0.22, +0.39]	+0.21 [+0.14, +0.26]
air	+0.49 [+0.35, +0.65]	+0.17 [+0.07, +0.25]

Table S16. The same contrast within strata of the true hidden share. Where the true window is observed an aligned outage has zero error by identity, so the first column restates the size of the false peak; a blank cell means one side of the contrast is empty. Source: `misalignment_within_strata.csv`.

reconstructor	true share 0	true share (0,1)	true share 1
climatology	+0.55 [+0.38, +0.75]	+0.17 [+0.10, +0.23]	+0.10 [−0.39, +0.44]
smoothed	+0.54 [+0.34, +0.78]	+0.16 [+0.07, +0.24]	+0.24 [−0.25, +0.44]
interpolation	+0.25 [+0.15, +0.32]	+0.10 [+0.04, +0.14]	—
donor	+0.30 [+0.17, +0.39]	+0.20 [+0.14, +0.26]	+0.16 [+0.08, +0.23]
air	+0.44 [+0.34, +0.52]	+0.18 [+0.07, +0.29]	−0.35 [−0.79, −0.09]

Table S17. Length effect (90 minus 30 days, °C) at a true hidden share of at least 0.9 and at least 0.999, by estimator. "Within station-years" keeps a station-year only when it holds both lengths above the cut-off (48 and 11 station-years); the "network subsets" estimator needs both lengths only somewhere in the same network and discards the pairing. Source: the 2026-10-14 mechanism run, `crossed_effects.csv`.

reconstructor	≥ 0.9, within station-years	≥ 0.9, network subsets	≥ 0.999, within station-years	≥ 0.999, network subsets
climatology	+0.12 [+0.05, +0.18]	+0.16 [+0.06, +0.25]	+0.08 [+0.03, +0.14]	+0.03 [−0.17, +0.22]
smoothed	+0.14 [+0.08, +0.20]	+0.20 [+0.12, +0.28]	+0.14 [+0.09, +0.18]	+0.10 [−0.05, +0.26]
interpolation	+0.82 [+0.63, +1.00]	+1.02 [+0.84, +1.19]	+1.12 [+0.75, +1.45]	+1.16 [+1.00, +1.33]
donor	+0.05 [+0.02, +0.08]	+0.12 [+0.04, +0.20]	+0.01 [−0.01, +0.05]	−0.09 [−0.24, +0.06]
air	+0.12 [+0.07, +0.18]	+0.22 [+0.10, +0.33]	+0.12 [+0.06, +0.16]	+0.13 [+0.03, +0.22]

Table S18. Mean absolute error (°C) by the temperature gap between the best 30-day window and the best window disjoint from it, crossed panel. The relation is non-monotone and flat for donor regression. Source: the 2026-10-14 mechanism run, `competing_peak.csv`.

gap to second window (°C)	outages	climatology	smoothed	interpolation	donor	air
[0,0.25)	189	0.54	0.53	0.21	0.19	0.35
[0.25,0.5)	342	0.33	0.31	0.24	0.19	0.23
[0.5,1)	477	0.41	0.37	0.32	0.17	0.27
[1,2)	594	0.42	0.41	0.47	0.21	0.39
≥2	297	0.56	0.60	0.68	0.20	0.54

Table S19. The composition rule and the realised unscreened failure rate in each real-calendar network (%). The prediction uses designed outages only; the three observed columns are the target-only, partial and all-donor failure scenarios. A network exceeds the 5% cap in two scenarios (the first two rows) and in four at the all-donor level. Source: `r2_05_screening_need/real_calendar_by_network.csv`.

network	groups	target station-years	overlap > 0.5 (%)	median gap (d)	predicted	observed, target only	observed, partial	observed
chmi_morava	6	6	100	364	47.8	66.7	66.7	
nve_basin_26	12	3	50	48	19.5	50.0	50.0	
nve_basin_88	27	6	11	39	5.9	3.7	3.7	
usgs2_huc2_08	95	6	0	9	1.4	1.1	0.0	
gkd_bayern_iller	9	6	0	2	0.8	0.0	0.0	
huc8_03110206	27	6	22	5	10.8	0.0	0.0	
usgs2_huc2_15	12	3	0	6	0.2	0.0	0.0	
usgs2_huc4_0601	24	6	0	4	0.8	0.0	0.0	
usgs2_huc4_1901	24	3	0	8	0.4	0.0	0.0	
usgs2_huc6_020502	45	6	0	5	0.8	0.0	0.0	
usgs2_huc6_070900	36	6	0	4	0.2	0.0	0.0	
usgs2_huc6_071300	38	4	0	10	1.6	0.0	0.0	
usgs2_huc6_120301	33	3	9	13	5.2	0.0	0.0	
usgs2_huc6_180400	30	3	0	7	1.0	0.0	0.0	

Table S20. The rule's decisions over the 14 networks, by failure scenario, with an exact binomial 95% interval for accuracy. "Missed" counts networks that needed a screen and were told they did not. Source: `real_calendar_rule_confusion.csv`.

scenario	networks	needing a screen	missed	screened unnecessarily	accuracy (%) [95% interval]
all donor failure	14	4	0	1	93 [66, 99.8]
partial donor failure	14	2	0	3	79 [49, 95.3]
target only	14	2	0	3	79 [49, 95.3]

Table S21. Excess refusal of true hot years (20 °C) over the hot-year-blind null: retention of all records minus retention of true hot years, in points, with group-clustered 95% intervals at each fold seed. Negative means hot years are refused less often than records in general. Source: `r2_04_hot_year_null/hot_year_selection_null.csv`.

panel	rule	seed 101	seed 202	seed 303
external	fitted risk	−1.0 [−1.6, −0.3]	−1.0 [−1.7, −0.3]	−0.8 [−1.4, −0.1]
external	same model, no support	−1.3 [−2.1, −0.4]	−1.1 [−1.9, −0.3]	−0.9 [−1.6, −0.2]
external	certificate bound	−1.0 [−1.7, −0.3]	−0.9 [−1.6, −0.2]	−1.0 [−1.7, −0.3]
external	peak-window rule	−0.3 [−0.9, +0.2]	−0.2 [−0.7, +0.4]	−0.1 [−0.6, +0.5]
external	gap-length rule	−0.8 [−2.2, +0.0]	−0.8 [−2.2, +0.1]	−0.8 [−2.2, +0.0]
development	fitted risk	−0.1 [−2.2, +1.4]	+0.3 [−1.7, +1.7]	+0.4 [−1.7, +1.8]
development	same model, no support	−0.1 [−2.0, +1.2]	+0.3 [−1.7, +1.7]	+0.2 [−1.7, +1.6]
development	certificate bound	−1.8 [−3.4, −0.2]	−1.9 [−3.6, −0.4]	−1.8 [−3.4, −0.2]
development	peak-window rule	−0.7 [−1.9, +0.3]	−0.7 [−1.9, +0.2]	−0.7 [−2.0, +0.3]
development	gap-length rule	−1.9 [−3.4, −0.3]	−2.3 [−3.9, −0.7]	−2.3 [−3.9, −0.6]

Table S22. Retention, refusal of true hot years, wrong hot-year judgements (% of candidates) and the break-even abstention cost c_{A}^{*} at $\rho = 1, 5, 10$ (the cost of a missed hot year relative to a false alarm), fold seed 101, 20 °C. A screen beats reconstructing everything only if one abstention costs less than c_{A}^{*} wrong judgements.

panel	rule	retention (%)	hot refused (%)	wrong (%)	c_{A}^{*}, $\rho = 1$	$\rho = 5$	$\rho = 10$
external	fitted risk	74.5	24.5	0.45	0.046	0.159	0.299
external	same model, no support	73.5	25.2	0.38	0.047	0.161	0.303
external	certificate bound	72.9	26.2	0.44	0.044	0.152	0.286
external	peak-window rule	66.5	33.2	0.49	0.034	0.129	0.249
external	gap-length rule	39.7	59.5	0.19	0.024	0.088	0.168
development	fitted risk	70.3	29.6	0.42	0.062	0.232	0.445
development	same model, no support	70.7	29.2	0.46	0.061	0.229	0.439
development	certificate bound	72.8	25.4	0.61	0.061	0.233	0.449
development	peak-window rule	60.0	39.3	0.35	0.048	0.176	0.337
development	gap-length rule	31.0	67.1	0.14	0.031	0.114	0.219

Table S23. Retention at the 5% cap by rule and panel (mean of three fold seeds, with the seed range in parentheses), and the realised failure rate among admitted records. The reference arm reconstructs every record and is included for scale; it does not satisfy the cap on the designed panels.

rule	external retention	external risk	development retention	development risk
gap-length	0.396	0.018	0.305	0.024
peak-window	0.667	0.033	0.604	0.038
certificate bound	0.729	0.032	0.727	0.036
fitted risk	0.740	0.031	0.704	0.030
reconstruct everything	1.000	0.153	1.000	0.149

Table S24. Hot-year composition, external panel, 20 °C, fold seed 101, case-weighted. Shares are fractions. The unique-station-year weighting is reported in the text and leaves the ordering unchanged. "Refused hot" is the share of true hot years left out of the archive; "missed hot" is the share the admitted reconstruction places below the threshold.

rule	p_{full}	p_A	selection	total (95%)	refused hot	missed hot
peak-window	0.857	0.861	+0.004	+0.003 [−0.005, 0.012]	0.332	0.003
fitted risk	0.857	0.868	+0.011	+0.007 [−0.002, 0.016]	0.245	0.004
certificate bound	0.857	0.868	+0.011	+0.007 [−0.002, 0.017]	0.262	0.004
gap-length	0.857	0.875	+0.018	+0.016 [−0.003, 0.046]	0.595	0.001
reconstruct everything	0.857	0.857	0.000	−0.006 [−0.013, 0.003]	0.000	0.013

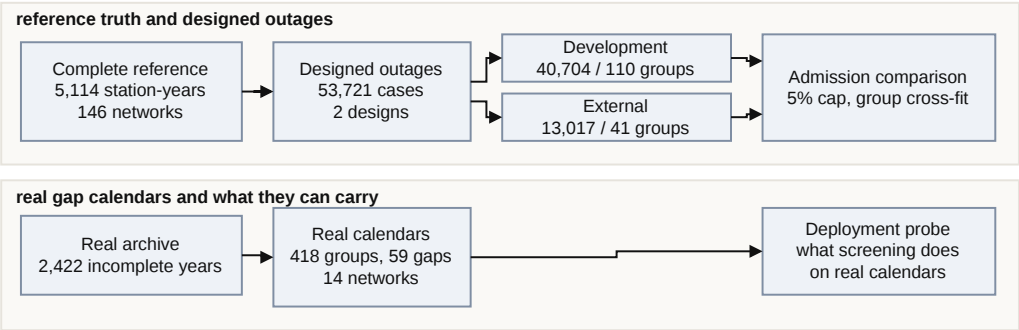


Figure S1. The evidence chain. Complete station-years are the reference truth; the development and external panels place stratified (single-run) and transplanted (real-geometry) outages and supply the admission comparisons; the real archive supplies gap geometry only, because it has no complete counterpart; and the 418 real-calendar groups, from 59 recorded gaps on 67 target station-years, supply the deployment probe. The separately generated crossed panel (1,899 outages on 211 station-years) that identifies the length and position effects is not drawn. Numbers give cases (and, for the panels, independence groups). The external panel is already opened, so its comparisons are cross-fitted within it rather than a sealed confirmation. Schematic source: `s1_study_design.drawio` (draw.io; see the Open Research statement of the article).